



SAFETY Department
SIMHADRI

Ref: SMTTPP/SAFETY/2022-23/NOTE/9855

Date:

05/04/2022(11:54:45)

Subject: Approval for LMI on Safety Monitoring, Measurement and Improvement

Subject:- Approval for LMI on Safety Monitoring, Measurement and Improvement - LMI-OPS-SAFT-006_Rev0,

Ref: COS-ISO-00-OD/OPS/SAFT/006, Rev. No.: 0 Feb 2022

Location Management Instructions made on ***Safety Monitoring, Measurement and Improvement***

put up for approval

Encl:-

1. OD-OPS-SAFT-006_Rev0_Safety Monitoring, Measurement and Improvement
2. LMI-OPS-SAFT-006_Rev0_Safety Monitoring, Measurement and Improvement

Submitted by:

पुली सदाशिव रेड्डी - वरिष्ठ प्रबंधक

Puli Sadashiva Reddy - 062135 (SENIOR MANAGER ,SAFETY) ,Simhadri
(PSADASHIVAREDDY@NTPC.CO.IN, 9425823806)

On: 05/04/2022(11:54:45) **Ref:** SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

चिप्पासत्य नारायण - अपर महाप्रबंधक

Chippa Satya Narayana - 031083 (ADDL. GENERAL MANAGER ,SAFETY)
(CSNARAYANA@NTPC.CO.IN, 9491876071) ,Simhadri

On:05/04/2022(16:15:42) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

आई एल रेड्डी - अपर महाप्रबंधक

ITIKYALA LAXMA REDDY - 032235 (ADDL. GENERAL MANAGER ,MTP)
(ILREDDY@NTPC.CO.IN, 9440918055) ,Simhadri

On:05/04/2022(18:14:46) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted for approval pls

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
(JHIMLIDHAR@NTPC.CO.IN, 9650998085) ,Simhadri

On:06/04/2022(11:37:07) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

नीलाजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:07/04/2022(12:49:24) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

शेषागिरी राव सत्तिरजु - अपर महाप्रबंधक

Seshagiri Rao Sattiraju - 005797 (ADDL. GENERAL MANAGER ,ENV. MGT.)
(SSRAO01@NTPC.CO.IN, 9440918215) ,Simhadri

On:11/04/2022(10:44:42) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

ASIS KUMAR MUKHOPADHYAY - 004586 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)

(AKMUKHOPADHYAY@NTPC.CO.IN, 9434038935) ,Simhadri

On:11/04/2022(12:48:27) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

एन वारा प्रसाद - अपर महाप्रबंधक

Nagalla Vara Prasad - 003913 (ADDL. GENERAL MANAGER ,FGD)
(NVARAPRASAD@NTPC.CO.IN, 9424142942) ,Simhadri

On:11/04/2022(15:43:43) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Reviewed

Submitted by:

एन वेणुगोपाल - अपर महाप्रबंधक

N Venugopal - 004781 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(VENUGOPALN@NTPC.CO.IN, 9412715649) ,Simhadri

On:12/04/2022(11:37:07) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

Amulya Kumar Behera - 097433 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(AMULYABEHERA@NTPC.CO.IN, 9425534368) ,Simhadri

On:12/04/2022(12:05:11) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

वि सिवा प्रसाद - अपर महाप्रबंधक

V Siva Prasad - 031702 (ADDL. GENERAL MANAGER ,STORES)
(VSIVAPRASAD@NTPC.CO.IN, 9440918096) ,Simhadri

On:13/04/2022(11:22:01) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

पुली सदाशिव रेड्डी - वरिष्ठ प्रबंधक

Puli Sadashiva Reddy - 062135 (SENIOR MANAGER ,SAFETY)
(PSADASHIVAREDDY@NTPC.CO.IN, 9425823806) ,Simhadri

On:18/04/2022(09:52:53) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:18/04/2022(16:27:56) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Pl see

Submitted by:

जीजीके मूर्ति - अपर महाप्रबंधक

Gumma Gopala Krishna Murthy - 004468 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(GGKMURTHY@NTPC.CO.IN, 9425224918) ,Simhadri

On:18/04/2022(16:48:24) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,OPERATION)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:18/04/2022(17:41:48) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

अर्जुन प्रदीप नायर - अपर महाप्रबंधक

Arjun Pradeep Nair - 004419 (ADDL. GENERAL MANAGER ,TECH SERVICES)
(APNAIR@NTPC.CO.IN, 9403964693) ,Simhadri

On:19/04/2022(09:07:08) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

अक्षय कुमार दास महापात्र - अपर महाप्रबंधक (चिकित्सा)

Akshaya Kumar Das Mohapatra - 032538 (ADDITIONAL GENERAL MANAGER (MEDICAL)
,PHYSICIAN)

(AKDASMOHAPATRA@NTPC.CO.IN, 9437572538) ,Simhadri

On:20/04/2022(11:00:38) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

कासा वेंकटेश्वर प्रसाद - अपर महाप्रबंधक

Kasa Venkateswara Prasad - 006486 (ADDL. GENERAL MANAGER ,REGIONAL LEARNING
INSTITUTE)

(KVPRASAD02@NTPC.CO.IN, 9650997388) ,Simhadri

On:20/04/2022(12:48:17) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

सन्जित मजूमदार - अपर महाप्रबंधक

Sanjit Majumdar - 005065 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(SMAZUMDAR@NTPC.CO.IN, 9650990065) ,Simhadri

On:20/04/2022(12:51:33) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL HAND.- OPN)
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

On:21/04/2022(15:39:00) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

प्रसन्न कुमार के एस - अपर महाप्रबंधक

K S Prasanna Kumar - 004504 (ADDL. GENERAL MANAGER ,IT)
(PRASANNAKMR@NTPC.CO.IN, 9428822389) ,Simhadri

On:25/04/2022(11:00:03) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

रूमा डे शर्मा - अपर महाप्रबंधक

Ruma De Sharma - 006923 (ADDL. GENERAL MANAGER ,HR)
(RDSHARMA@NTPC.CO.IN, 9650999329) ,Simhadri

On:25/04/2022(11:13:16) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

LMI reviewed and forwarded for approval

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:26/04/2022(18:21:45) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

Vishwa Mohan Choudhary - 003932 (GENERAL MANAGER ,FUEL MANAGEMENT)
(VMCHOUDHURY@NTPC.CO.IN, 7588691232) ,Simhadri

On:27/04/2022(10:27:57) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Please go through the proposed LMI and circulate widely among all EMD group heads. Also, keep a soft copy of the proposed LMI for your reference also. Please mark the file to AGM(Safety) for correction in page 3 (for adding names of EMD and Operation Heads).

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:29/04/2022(18:34:32) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Review and comment , if any

Submitted by:

धर्मेंद्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:02/05/2022(17:01:37) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Awareness of Safety Policy is being communicated to all employees as well as contract workers.

Topics & discussions of daily safety peptalks are being recorded in SURAKSHA app.

Safety pledge is being administered in daily TBT and Safety peptalks.

All safety signs & posters related to electrical hazards are posted at respective locations.

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

Makala Nageswara Rao - 007521 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(MAKALANAGESWAR@NTPC.CO.IN, 8331016103) ,Simhadri

On:04/05/2022(17:11:21) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

बाला बालाजी आकुला - वरिष्ठ प्रबंधक

BALA BALAJI AKULA - 009145 (SENIOR MANAGER ,ELECT MAINT)
(ABALABALAJI@NTPC.CO.IN, 8332959245) ,Simhadri

On:05/05/2022(09:21:47) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

KRISHNA BHESETTY - 009044 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(BKRISHNA@NTPC.CO.IN, 9493546044) ,Simhadri

On:09/05/2022(17:35:42) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Awareness of Safety Policy is being communicated to all employees

Daily safety peptalks are being conducted and recorded in SURAKSHA app.Safety pledge is being administered Safety peptalks.All safety signs & posters related to electrical hazards are displayed in respective loactions of electrical equipments.

Submitted by:

वी वी रमेश - उप महाप्रबंधक

Vurumu Venkata Ramesh - 007523 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(VVENKATARAMESH@NTPC.CO.IN, 9493546033) ,Simhadri

On:10/05/2022(10:14:48) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

1. Contractor Safety Assessment and Finding and record keeping on monthly basis, necessary checklist preparation in progress and will be incorporated in all future maintenance contracts, existing running contracts checklist preparation under progress.

2. High risk hotspot are being identified and safety performance improvements of EMD-SYD,MUW,ESP areas in progress.

Submitted by:

रवि वर्मा कनुमुरी - उप महाप्रबंधक

Ravivarma Kanumuri - 008001 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(RAVIVARMAK@NTPC.CO.IN, 9493174808) ,Simhadri

On:11/05/2022(09:49:31) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

धर्मेंद्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:11/05/2022(16:45:26) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

Submitted by:

चिप्पासत्य नारायण - अपर महाप्रबंधक

Chippa Satya Narayana - 031083 (ADDL. GENERAL MANAGER ,SAFETY)
(CSNARAYANA@NTPC.CO.IN, 9491876071) ,Simhadri

On:12/05/2022(09:30:25) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

LMI reviewed and forwarded for approval

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

Rajasekhar Pala - 005146 (ADDL. GENERAL MANAGER ,OPN (MAIN PLANT))
(PRAJSEKHAR@NTPC.CO.IN, 8331015741) ,Simhadri

On:13/05/2022(17:53:45) **Ref:**SMTPP/SAFETY/2022-23/NOTE/9855

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
(RKJHA02@NTPC.CO.IN, 9415342046) ,Simhadri

On:13/05/2022(20:40:57) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

Girish Chandra Choukse - 002907 (GENERAL MANAGER ,HEAD OF PROJECT)
(GCCHOUKSE@NTPC.CO.IN, 9425178006) ,Simhadri

On:14/05/2022(10:33:59) **Ref:**SMTTPP/SAFETY/2022-23/NOTE/9855



SIMHADRI

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

Location Management Instruction – LMI-OD-OPS-SAFT/006

Revision: 00, Dt: FEB - 2022

SAFETY MONITORING
MEASUREMENT & IMPROVEMENT
AT
SIMHADRI SUPER THERMAL POWER STATION

Approved for Implementation by.....
HEAD OF PROJECT (Simhadri)

Date: __.03.2022

SAFETY DEPARTMENT

TITLE OF LMI:		Safety Monitoring, Measurement and Improvement at simhadri super thermal power station					
LMI No:		LMI-OD-OPS-SAFT/006					
ISSUE NO	REVISION NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
01	00	.03.22	All clauses are new as the document is being issued for the first time.	Safety Officer	ALL HODs	GM(O&M)	HOP

NTPC LIMITED**SIMHADRI SUPER THERMAL POWER STATION****LMI APPROVAL DOCUMENT**

Title of LMI:	Safety Monitoring, Measurement and Improvement at simhadri super thermal power station		
LMI No.:	LMI-ODN-OPS-SAFT/006		
Issue No. : 1	Revision: 0	Date : .03.2022	

	Name Sri.	Designation	Signature
Prepared by	P Sada Shiva Reddy	Sr. Manager (Safety)	
Checked by	Ch Satyanarayana	AGM (Safety)	
	I Laxma Reddy	AGM(MTP)	
	Jhimli Dhar	AGM (EEMG)	
	Nilanjan Mondal	AGM(AHP)	
	S Sesha Giri Rao	AGM (EMG)	
	Asis Kumar Mukhopadhyay	AGM(OFS & CIVIL)	
	N Vara Prasad	AGM (FGD)	
	N Venugopal	AGM (Chemistry)	
	Amulya Kumar Behera	AGM(FT&FC)	
	V SivaPrasad	AGM(C&M Stores)	
	Om prakash	AGM (BMD)	
	G G K Murthy	AGM (TMD)	
	Arjun Pradeep Nair	AGM (TS & FQA)	
	A K Das Mohapatra	CMO	
	K V Prasad	HR-RLI	
	Sanjit Majumdar	AGM-FM	
	K S Prasanna Kumar	AGM(IT)	
	Ruma De Sharma	AGM (HR)	
	T R Rajagopalan	AGM (C&I)	
	Vishwa Mohan Choudhary	GM(FM)	
	Mathew Eipe Kooror	GM(Maint.)	
Reviewed by	R K Jha	GM(OPN)	
Approved for Issue by	G C Choukse	GROUP GENERAL MANAGER (S)	
Date of Issue			

Table of Contents

1.0 SUMMARY	2
1.1 Superseded Document.....	2
1.2 Scope	2
1.3 Review	2
2.0 INTRODUCTION.....	2
3.0 RECORDS OF REVISION	2
4.0 PROCESS.....	2
4.1 ROLES & ACCOUNTABILITY.....	2
4.2 REQUIREMENTS	3
4.2.1 QUANTITATIVE MONITORING	3
4.2.2 QUALITATIVE MONITORING	3
4.2.3 REACTIVE MONITORING & MEASURES.....	5
4.2.4 CONTINUAL IMPROVEMENT IN SAFETY	5
4.3 AUDIT	6
5.0 RECORDS TO BE MAINTAINED	7
6.0 ASSOCIATED DOCUMENTS.....	7

SUMMARY

Superseded Document	Not Applicable	
Scope	NTPC Simhadri Super Thermal Power Station	
Review	Review Responsibility	Safety dept.
	Review Frequency	2 yearly basis, or as and when required, whichever is lower
	Review Approval	Head of Project

INTRODUCTION

- a. NTPC Top Management is committed to continual improvement of its safety system. One critical aspect of this commitment is that safety performance is monitored, measured, and analyzed such that improvement can sustain on a continual basis. The Objective of this procedure is to verify that the standards and minimum requirements as stated in NTPC safety framework are implemented and followed at the location; ensure that the defined requirements are measured to assess their effectiveness; and a systematic performance feedback mechanism is in place for periodic reviews.
- b. Towards achievement of this objective, the location shall implement and maintain procedures to monitor and measure Occupational Health & Safety performance on a regular basis, and the extent to which the safety & occupational health objectives are met. The measuring and monitoring should primarily focus on proactive measures to effect safety and occupational health performance improvement and injury reduction, but also need to use reactive measures.
- c. *The reference OD does not contain elements that can vary as per location. Hence no modification or, adaptation for location is expected in this LMI.*

3.0 RECORDS OF REVISION

Not Applicable

4.0 PROCESS

4.1 ROLES & ACCOUNTABILITY

i. Head of the Project/Stations (HOP) is accountable for:

- a. Ensure a system is in place for safety performance monitoring, measurement.
- b. Ensure a system is in place for analysis of safety performance data.
- c. Ensure a system is in place for continual improvement of safety performance.
- d. Ensure a system is in place for taking feedback from employees, outsourced persons.
- e. Encourage and ensure full participation in safety perception surveys.
- f. Ensure a system of cross-function team audits

ii. Head of Function/Section is accountable for:

- a. Schedule of function/section safety monitoring.
- b. Work towards improving risk controls in identified high-risk hot spots for safety performance improvement of respective area.

iii. Site Safety Coordinator is accountable for:

- a. Prepare and circulate, annual safety-related audits calendar in consultation with CC-Safety.
- b. Ensure to include in agenda of Safety Committee Meetings: safety performance report, feedbacks, incident recall case studies, safety improvement programs.
- c. Analysis of safety performance data and generating relevant metrics.
- d. Identify hot spots for safety performance improvement of respective area.
- e. Track and review periodically with HOP the safety performance metrics of the location.
- f. Update periodically central repository safety performance metrics of the location.

iv. Location HR Head is accountable for:

- a. Arrange for training on Internal Auditors.

v. Where operations and maintenance functions are outsourced:

For projects where operations and maintenance functions are outsourced to third- parties, the designated Person accountable for the performance of such a project, wherever he/she is located, shall be accountable for the above roles. In this scenario, the safety awareness, communication, and promotion shall be for the location outsourcing agency as a whole and shall be included as an agenda item in every review of such project's performance.

4.2 REQUIREMENTS

The location shall implement and maintain procedures to monitor and measure Occupational Health & Safety performance on a regular basis, to monitor the extent to which the safety & health objectives are met and the effectiveness of risk controls. The measuring and monitoring should primarily focus on proactive measures to effect improvement in safety and occupational health performance and reduce injury incidence; and would also need to use reactive measures.

Proactive monitoring includes qualitative, and quantitative measures.

4.2.1 QUANTITATIVE MONITORING

Quantitative measures include using equipment to monitor/measure health and safety conditions (e.g. noise meters, toxic gas detection equipment). Such equipment should be maintained and calibration traceable to applicable international or national standards.

Recording of data and results of monitoring and measurement, sufficient to facilitate subsequent corrective action and preventive action analysis shall be done.

Quantitative monitoring can be included as part of Qualitative Monitoring. **(Responsibility: Chemistry)**

4.2.2 QUALITATIVE MONITORING

Qualitative monitoring can be in the form of compliance checks with legal, regulatory safety & health requirements, workplace safety inspections, evaluation of the effectiveness of safety & health training through planned job observations/hazardous task observations, use of behavior based observation, use of perception surveys to evaluate safety culture, completion of legally required and other inspections as scheduled, work activity assessments, internal & external audits.

(Responsibility: All departments)

4.2.3 Monitoring by Functions

These include scheduled assessments of aspects of the respective section/department, work activity assessments, planned job observations/hazardous task observations and shall comply with the requirements laid out by the OGN - Periodic Safety Assessments (Internal to Project).

(Responsibility: All departments)

4.2.4 Monitoring Safety Performance of Contractors of the Section

The respective HOS shall ensure that all contractors and their worksites shall be subjected to scheduled health & safety assessments including planned job observations/hazardous task observations and behavioral aspects. The respective HOS must: decide what constitutes adequate monitoring, based upon knowledge of the work; take steps along with the Area Engineer to ensure that contractors work as specified in the job plan/construction plan. A table of high-risk activities with risk-control measures that would be ensured by the contractor shall be maintained and monitored by the respective Area Engineer/HOS.

The responsible Area Engineer or HOS should conduct such monitoring assessments. Details of such inspections shall be captured in the centralized system and included in periodic reports by the respective HOS. HOSs shall ensure that cognizant unsafe acts especially gross negligent behavior results in exemplary penal action. The respective HOS shall monitor such schedules regarding completion ratio, findings not closed, missed site visits etc.

(Responsibility: All departments)

Safety Monitoring by Cross-Functional Team of the Project

These shall comply with the requirements laid out under the OGN - Periodic Safety Assessments (Internal to Project).

Internal Audits

These are done by team(s) from outside the project but, within the company, based on requirements as decided by the Corporate Safety function. Such audits can be included as part of other compliance audits or assessments like technical compliance audits. The location shall include the scheduled internal audits in the location's annual calendar of audits. The respective Site Safety Coordinator shall monitor such schedules regarding audit completion ratio, findings closed ratio, missed audits etc.

The findings of all internal audits shall also be captured in a central system like SAP for retrieval, follow-up and closure. Site Safety Coordinator shall be responsible to communicate, follow-up, monitor, and provide updates on the same. Monitoring of the internal audit findings closure status shall be part of periodic reports.

External Audits

The location shall have a system in place to ensure that external safety audits are carried out, as required by legislation, statutes, regulations, or other requirements. The location shall include the scheduled external audits in the location's annual calendar of audits. The respective Site Safety Coordinator shall monitor such schedules regarding audit completion ratio, findings closed ratio, missed audits etc. The findings of external audits shall be reported in a central system like SAP for retrieval, follow-up and closure. Site Safety Coordinator shall be responsible to communicate, follow-up, monitor, and provide updates on the same. Monitoring of the external audit findings closure status shall be part of periodic reports.

Monitoring Safety Culture - Perception Surveys

Perception Survey is a good tool to understand and assess the prevailing safety culture at that point in time and the way it is changed or responded to interventions and initiatives. Use of periodic

perception surveys to evaluate safety culture shall be based on guidance from corporate functions or can be initiated by the location based on any change at the location. The HOP shall encourage and ensure full participation in all such perception surveys.

REACTIVE MONITORING & MEASURES

Reactive monitoring measures include: occurrences and rates of incidents/ill health, lost time incident/ill health rates, actions required following assessments by regulatory agencies, actions following receipt of comments or complaints from interested parties. Monitoring statistical data compiled and analyzed includes:

- actual man-hours worked,
- number of near misses (including numbers of unsafe conditions, unsafe acts),
- number of fatalities,
- number of reportable incidents,
- number of occupational illness and diseases,
- number of high-consequence injuries (LTI of more than two months),
- non-work related injuries within the location's defined premises,
- incidence rate,
- frequency rate,
- Disabling injury incident rate (DIIR) – *calculated as per NOSA norms*,
- Disabling injury frequency rate (DIFR) – *calculated as per NOSA norms*,
- number of days lost for each incident case,
- injured part of body for each injury case,
- partial permanent disability (% loss in hearing capability),
- compensation pay out,
- damage loss,
- incident causes/types.

Unless explicitly mentioned otherwise, calculations methodology adopted by the location, shall follow the following sequence of priority:

- regulations/statutes applicable to the location, like State Factories Act;
- IS-3786;
- other international standards/norms.

CONTINUAL IMPROVEMENT IN SAFETY

The location shall have in place a system of improvement in safety performance based on analysis of reactive measures.

Data Analysis

Data analysis is an essential tool to improve safety performance by identification of hot spots within the location where more vigilance would be useful, etc. Data shall be made available in a central system by the site safety coordinator to facilitate reports and analysis. The Site Safety Coordinator shall be the nodal point for data acquired through monitoring and measurement to assess the trends in performance and effectiveness of risk controls in the location's processes, to identify (with support of relevant HOD/HOS) root causes of non-conformances, potential new hazards, and evaluate where improvements can be made.

Incidents Analysis

Incidents analysis offers the location one of the best prospects for identifying opportunities to

improve and to raise the overall safety & health awareness in the workplace. Information obtainable by means of incident analysis include: how the accidents occur, where the accidents occur, nature of the injuries. Such analysis facilitates to identify hot spots in the location, and to prioritize preventive action based on risk (higher numbers of incidents). Site Safety Coordinator shall be responsible for collating and analyzing incidents & incident investigation reports and audit findings to generate learnings to improve the processes.

Improvement Through Audits

The most effective use of the results of internal and external audits is including them in the location's safety performance improvement programs. Audits (Internal and external) reports are a structured way of improving safety performance through identification of preventive actions and opportunities for improvement. Findings of audits shall be factored by the Site Safety Coordinator and respective HOSs, in developing and implementing safety initiatives and safety performance programs.

Improvement Through Feedback

To improve its safety and health performance, the location shall actively solicit suggestions, verbal or, in writing, from employees, contractors and outsourced workers. Such feedback can be of two types. One that is initiated by the location, or the corporate function in the form of perception surveys; and the second case at individual level, where individual employees or outsourced workers give voluntary feedback through an established system.

Improving Safety Culture - Perception Surveys

Data from safety perception surveys initiated by corporate offices shall be analyzed at by such corporate office for identifying areas of improvement in safety performance. Data from safety perception surveys initiated at the location shall be analyzed by the Site Safety Coordinator, for identifying areas of improvement in safety performance.

Feedback – Employees

Employees are a great channel for suggesting improvements to the safety performance and culture at a location. The location shall facilitate employee safety reporting and feedback by establishing a system or linking with a central system. Facility to report anonymously shall be provided to employees.

Feedback - Outsourced Workers & Contractors

Outsourced workers as the frontline facing shortfalls in safety standards offer another channel for suggesting improvements to the safety performance and culture at a location. Consultation and participation of outsourced workers and contractors is a legal requirement in many locations. Site Safety Coordinator shall include such feedback as part of safety committee meeting agenda and cover it during the meeting, even if there was no feedback received during the period under review, and include the same in the minutes of the location's safety committee meetings. Facility to report anonymously shall be provided to contractors, and outsourced workers.

Data from employee, contractor, and outsourced workers shall be analyzed at the location by the Site Safety Coordinator for identifying areas of improvement in safety performance.

4.3 AUDIT

- a. The audit shall be a combination of verification of: system, records, implementation in the field and interviews.

- **System:** Auditors shall validate the evidence: audits calendar, employee & outsourced worker feedback systems, safety monitoring of contractors by respective HOS, as specified under [4.2](#).
- **Records:** Auditors shall include verification of all the records detailed under 5.0. This shall be included as part of the audit report.
- **Interviews:** Verify with at least five employees randomly selected from different functions on their awareness of feedback system; Verify with at least five outsourced workers randomly selected from different sections on their awareness of feedback system.
- b. In a project where NTPC has outsourced operational responsibility to third-parties, but continues, as the project developer, to have a legal obligation of ensuring a safe working environment, the designated person accountable for such a project shall be accountable for ensuring periodic audits. Records of such audits shall be maintained.

5.0 RECORDS TO BE MAINTAINED

Any record that is maintained as an electronic copy shall be in a format like secure PDF, that cannot be edited. Records shall be maintained as mentioned under [4.1](#) by respective roles.

HOP:

- Employee and outsourced worker feedback system;
- Budget provision for continual improvement of safety performance.

HOD/HOS:

- % section self-assessments completed in time, % self-assessment findings closed.
- Record of Monthly safety assessment of respective contractors.
- % contractor assessments completed in time, % assessment findings closed.

Site Safety Coordinator:

- Annual schedule of audits;
- % internal audits completed in time, % internal audit findings closed
- % external audits completed in time, % external audit findings closed
- Safety performance data analysis and metrics of the location;
- Identification of high risk areas based on trends of near-miss & incidents;
- Record of initiatives for continual improvement;
- Safety Committee meeting agenda to include feedback, performance metrics.
- Safety Committee Meeting minutes to include feedback, performance metrics.

HR:

- Annual training schedule inclusion of training of Internal Auditors.

6.0 ASSOCIATED DOCUMENTS

Not Applicable